



Coloring and fusing architectural sketches by combining a Y-shaped generative adversarial network and a denoising diffusion implicit model

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Abstract

Sketch-based architectural design tasks require considerable time and effort. However, developing an artificial intelligence agent to color and fuse architectural sketches is challenging due to the imaginative nature of the task and technical limitations. Here, we first introduce Y-shaped generative adversarial networks (GANs), which can color sketches on the basis of temporarily specified reference images by embedding an attention mechanism model that can match spatial and color information and by maintaining a type of triangular equilibrium state. Then, we develop practicable methods for coloring and fusing the given sketches simultaneously using a combination of our Y-shaped GANs and the denoising diffusion implicit model, yielding results with high quality and veracity. A novel interpolation process is developed as a necessary trick for realizing the fusion function. The testing results indicate that this method works better than the existing methods for the tasks of coloring and fusing architectural sketches.

1 | INTRODUCTION

Sketch-based architectural design tasks involve many objective and subjective decision-making processes that integrate science and art to achieve better living conditions. In the early stage of design research, sketching was commonly regarded as an effective means of (1) developing early-stage ideas, integrating expert knowledge, balancing design goals, and solving unexpected problems; (2) pursuing innovations and preserving transient results in a step-by-step manner during the conceptional stages of design; (3) describing the structures of buildings; and (4) balancing personal viewpoints as a type of design language. During the design procedure, sketches are modified

by many basic patterns, such as by (1) revising specific portions or integrating entire sketches conceived by different designers, (2) using additional architectural images as references for optimization, and (3) drawing color renderings based on sketches for further comprehensive evaluations. In general, the sketch-based architectural design process usually has an iterative and exploratory nature (Buxton, 2007; Kowaltowski et al., 2010; Schon, 1984; Schütze et al., 2003).

During conceptual design, architects may utilize the existing design works as references when conceiving new designs, or even attempting to break free of the traditional notions of architecture. However, structural designers mainly use optimization methods when conceiving the



structures of buildings (e.g., Kociecki & Adeli, 2014, 2015). As such, architectural design assisting artificial intelligence (AI) agents usually have specific tendencies in their main functions in relation to different design stages. Typically, the generation networks are more suitable for conceptual design, and the optimization algorithms are more suitable for conceiving better architectural structures.

The existing studies have conceived many AI methods for assisting sketch-based architectural design tasks in fusing sketches by a generative adversarial network (GAN) or relying on a self-sparse module in GANs to avoid generating nearly white or mosaic sketches (Qian et al., 2021, 2023). However, these sketch data-based methods do not learn or utilize essential color information when fusing sketches. The style-transforming models, which work by balancing the content and style losses (e.g., Li & Wand, 2016; X. Wang & Gupta, 2016), are mainly used to generate artistic works and cannot fuse sketches. The U-Net-based GAN, CycleGAN (Zhu et al., 2017) and StyleGAN (Karras et al., 2019) architectures can be used to translate sketches into corresponding color images. However, these methods generate images with only a single style by balancing every case in the given dataset. Moreover, the fusion results of the sketches generated by these models are too blurry to use because these structures only focus on coloring tasks (see Figure 1).

Recently, many unconditional GANs have been developed as effective means of translating sketches into colored images (e.g., W. Chen & Hays, 2018; Lee et al., 2020; Sangkloy et al., 2017; S. Y. Wang et al., 2021). Self-reference face coloring GANs (Lee et al., 2020; S. Wu et al., 2022) have achieved state-of-the-art results in coloring facial sketches by using temporarily specified reference images and by integrating the U-Net architecture with attention mechanism (AM) modules. When investigating Lee's method (Lee et al., 2020), we discovered that it is applicable for coloring architectural images only by applying several modifications. (1) The coloring GAN should maintain a type of triangular equilibrium state (see Section 3.1) during the training processes, and only in this way, can the input sketch control the spatial characteristics of the output, while the input reference image determines the color. (2) Different from facial images, architectural images can be so different from each other both in terms of spatial and color-related aspects that the input images do not need twisting pretreatment, and the AM modules should delete the linear layers. In this research, we modify Lee's method into a Y-shaped GAN for coloring architectural sketches. Nevertheless, both Lee's GAN and our Y-shaped GAN are still weak in terms of fusing sketches for two reasons:

First, the more a GAN focuses on the coloring task, the worse it is at performing the fusion task if realized by implementing interpolation tricks (e.g., Karras et al., 2019; Radford & Metz, 2016). Taking the U-Net architecture as an example, the first-level feature maps of sketches are directly sent to the final layers by bridging links so that interpolating these feature maps only yields blurry images (see Figure 1).

Second, GAN-based models tend to generate curved edges or spherical surfaces instead of maintaining the linear, rectangular, and gravity-based structures of buildings. For example, generating a globular building based on two cuboid buildings is clearly illogical. Such deficiencies were commonly observed in previous GAN-based research (e.g., Qian et al., 2021, 2023). The discriminator in GANs can not improve this by using the "real" criterion.

Based on our tests and comparisons, we have observed that combining our Y-shaped GANs and diffusion models, such as denoising diffusion probabilistic models (Ho et al., 2020) or denoising diffusion implicit models (DDIMs; Song et al., 2020), can effectively solve the above problems. However, we have observed that a DDIM generates only nearly white results when it is directly trained on sketch data. Accordingly, we can first color architectural sketches using temporarily specified reference images via our proposed Y-shaped GAN to obtain a pair of color-similar images. Then, these images can be fused using a DDIM trained on colorful architectural images (see Figure 2). In this case, the DDIM works both as a generator and a deblurring network. The interpolation tricks are still effective if a sketch-color pair or two color images are interpolated before the encoding process.

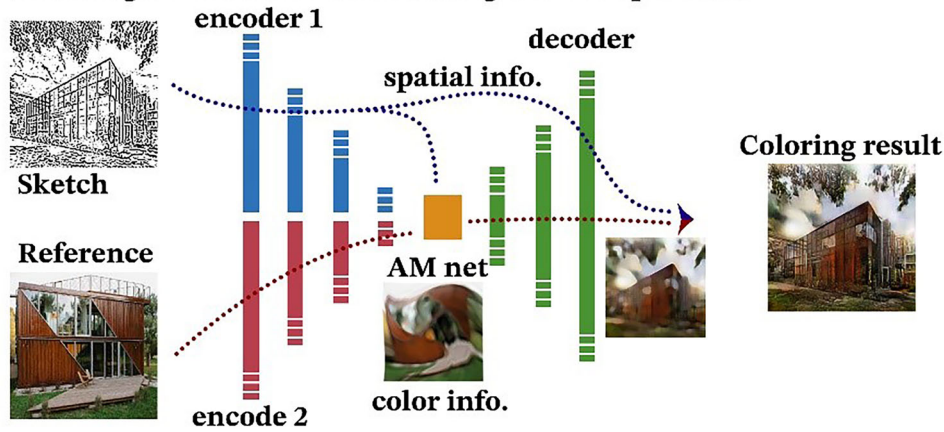
Our first contribution is conceiving effective methods for simultaneously coloring and fusing architectural sketches by using a combination of a Y-shaped GAN and a DDIM. Our second contribution provides potential approaches for consolidating the conceptual and structural design processes. The optimized structures can be conveniently visualized for evaluation.

The remainder of this paper is organized as follows. In Section 2, a literature review of computer-aided conceptual and structural design methods for buildings is summarized. The operational procedures of our novel methods are explained in Section 3, and we report the corresponding test results in Section 4. Finally, conclusions are presented in Section 5. The source codes are available at <https://github.com/abc637725/multi-headed-Y-GAN>, and an online test network of Y-shaped GAN is available at <http://www.y-gan.net/>.



FIGURE 1 Coloring and fusing architectural sketches with different methods. DDIM, denoising diffusion implicit model; GAN, generative adversarial network.

(a) Coloring architectural sketch using the Y-shaped GAN



(b) Coloring and fusing architectural sketches using the Y-shaped GAN and DDIM

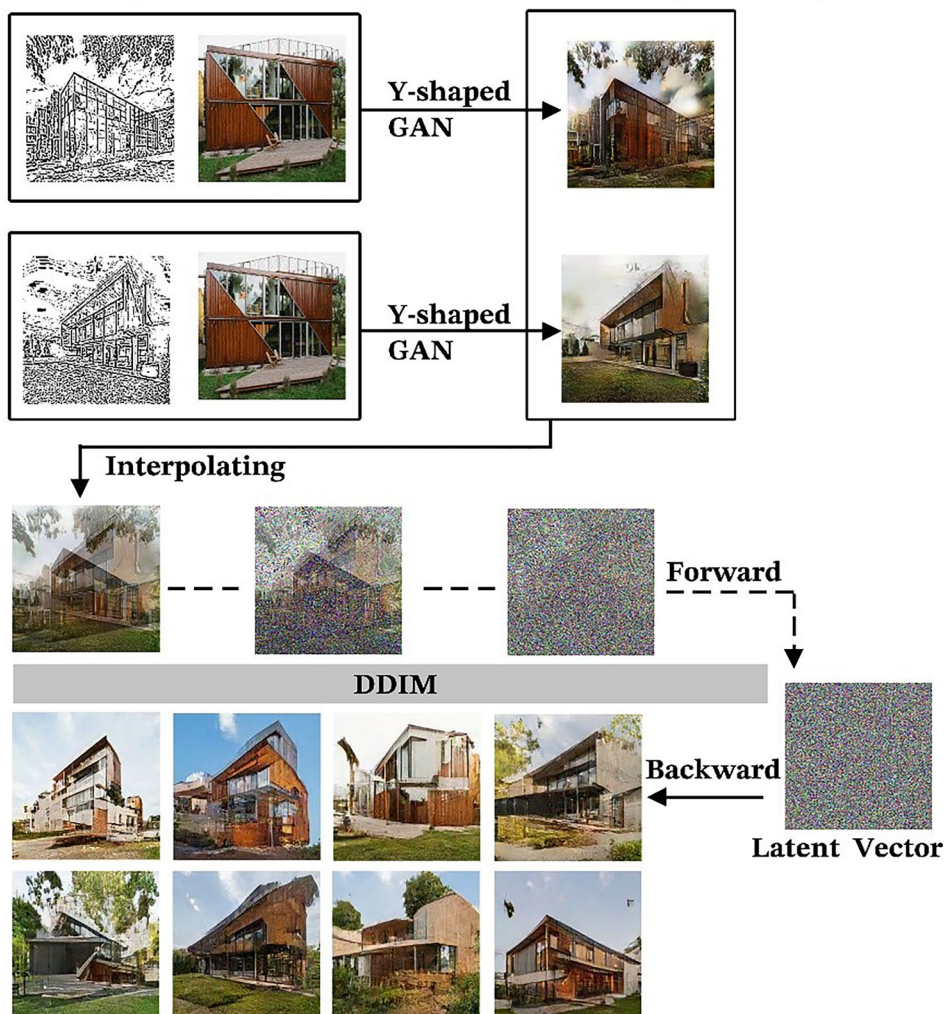


FIGURE 2 The methodology for coloring and fusing architectural sketches. AM, attention mechanism; DDIM, denoising diffusion implicit model; GAN, generative adversarial network.



2 | RELATED WORK

2.1 | The conceptual and structural designs of buildings

2.1.1 | Computer-aided structural design optimization

Aldwaik and Adeli indicated that the goal of structural optimization may be threefold: (a) automating the complex design process; (b) optimizing the performance of the constructed structure, such as its weight, strength, stiffness, stability, functionality, and even aesthetics; and (c) minimizing the total cost of the structure (Aldwaik & Adeli, 2014). Therefore, the optimization objects of a building structure are usually set to minimize key factors such as the cost and weight while constraining other factors with basic requirements. Long-term structural optimization approaches have been developed by utilizing newly proposed computational methods such as genetic algorithms (Adeli & Kumar, 1995), neural dynamics algorithms (Park & Adeli, 1997), and particle swarm algorithms (Plevris & Papadrakakis, 2011).

The structural design optimization process is usually scheduled after the conceptual design step, even though it is crucial for realizing the initial ideas. For example, the grid elements in roof structures should be rearranged within a given shape created by architects (Kociecki & Adeli, 2014, 2015). This procedure is established for many reasons. (1) Conceptual designers conceive solutions by simultaneously utilizing optimization thinking and divergent thinking, while structural designers mainly focus on satisfying objectives via optimization methods; and (2) designers in different phases cannot sufficiently exchange ideas in a limited amount of time.

2.1.2 | Deep learning methods for optimization, classification, and generation tasks

Deep learning models such as convolutional neural networks (CNNs; Liao et al., 2021), recurrent neural networks (B. Zhang et al., 2022), and many GANs-based approaches have been used in the domains of civil and engineering, such as for detecting structural damage (Gao et al., 2021; Maeda et al., 2021), avoiding vessel-bridge collisions (B. Zhang et al., 2022), estimating vehicle weights (Yu et al., 2021), imputing travel times (K. Zhang et al., 2021), and generating structural images of buildings (Liao et al., 2021).

The optimization equations in deep learning models, such as the L1 loss for generators, are mainly used for effectively training and utilizing the existing data. As a per-

tinuous method, self-supervised learning focuses on how to utilize unlabeled data (Rafiei et al., 2022). In this instance, the functions of satisfying traditional knowledge-based goals, such as minimizing the weights and costs of buildings, are realized by additional losses that are not essential for realizing data-based learning tasks.

Numerous deep learning methods have been developed for classification tasks. For example, many newly developed approaches have proven effective for solving classification problems (e.g., Alam et al., 2020; Liao et al., 2021; Pereira et al., 2020; Rafiei & Adeli, 2017). However, structural design works are more related to optimization and generation tasks than to classification tasks.

2.1.3 | Linking the conceptual and structural design processes via deep learning methods

Currently, effective strategies for simultaneously learning massive amounts of data and satisfying knowledge-based goals are crucial for assisting the structural design process via deep learning methods. One of the commonly used strategies is labeling the data in conditional generation processes. Another strategy is to conceive composite losses in deep learning architectures such as GANs to generate and optimize the structures of buildings. Theoretically, using these strategies is equivalent to using traditional optimizing methods such as neural dynamics algorithms. Even though, the AI agent still needs some methods for utilizing other information such as architectural features embedded in trained data.

2.2 | Coloring architectural sketches

In recent years, colorful design images have mainly been generated through modeling, rendering, cutting, and pasting via many image processing software programs, such as SketchUp, 3DS MAX, and Photoshop. However, these approaches all incur greater or lesser mechanical and time costs for design tasks. To address these problems, many AI agents have been developed to assist in architectural design based on various approaches, such as image retrieval, optimization-based, graph-based, and data-driven approaches.

2.2.1 | Image retrieval

The traditional sketch-based image retrieval methods are the synthesis techniques of Photosketcher (Eitz et al., 2011) and Sketch2photo (T. Chen et al., 2009). Photosketcher utilizes the bag-of-features approach to realize retrieval



from an image database, and Sketch2photo generates photorealistic pictures by means of selection, filtering, and blending methods. Novel text and sketch transformers, such as TASK-former (Sangkloy et al., 2022) and content-based models (Lu et al., 2022), use both text descriptions and sketches as input to retrieve more relevant images.

2.2.2 | Image-to-image translation via GANs

For sketch-coloring tasks, which can be treated as a class of image-to-image translation problems, the community has already taken significant steps in this direction with CNNs (LeCun et al., 1998), U-Nets (Ronneberger et al., 2015), and GANs (Goodfellow et al., 2014) as the basic models. For example, SketchGAN can synthesize plausible images from 50 categories, including motorcycles, horses, couches, and many others (W. Chen & Hays, 2018). The CycleGAN (Zhu et al., 2017) and StyleGAN (Karras et al., 2019) methods, as well as the Wasserstein GAN architecture (W-GAN, Arjovsky et al., 2017), can also be used to translate sketches into corresponding color images. However, these methods generate images with only a single style by balancing every case in the dataset, without using controllers during the translation processes.

In the academic field of architectural design, Huang and Zheng have generated apartment plans based on labeled images via GANs (Huang & Zheng, 2018), and ArchiGAN has been proposed to generate architectural floor plan images based on a series of Pix2Pix-GANs trained on paired datasets (Chaillou, 2019). These methods only generate unique corresponding results for the inputs.

2.2.3 | Controlling translation via conditional GANs (cGANs)

Based on the CNN, U-Net, and GAN architectures, many subsequently developed models focus on identifying interpretable mechanisms for translation among datasets so that the translation processes can be deliberately controlled by humans across several domains. Accordingly, a series of cGANs have been developed in which labeled images or one-hot vectors are utilized as controllers (Choi et al., 2018; Isola et al., 2017; R. Zhang et al., 2017). For example, TextureGAN was conceived for generating images of common items from hand-drawn sketches using texture images as the controllers (Xian et al., 2018). Another cGAN-based image synthesis system allows users to scribble sketches via sparse colorful strokes to generate images of cars, faces, and bedrooms (Sangkloy

et al., 2017). Recently, Zhou conditionally trained an architectural sketch-coloring GAN considering four stylistic variations (Zhou & Park, 2023).

Moreover, if the contents of the supplemental inputs have spatial relationships with each other, such as two nodes used to express the spatial layout of several rooms or a sentence used to describe text-to-image relationships, the corresponding cGANs can also be regarded as graph-based GANs or semantic processing GANs (e.g., Sangkloy et al., 2022). Such methods have been widely used to assist in architectural layout design tasks (e.g., Hu et al., 2020; Huang & Zheng, 2018; Nauata et al., 2021; W. Wu et al., 2019). In addition, different from classic facial image generation tasks, the image data of buildings can be so different from each other that developing conditional-based networks for 3D buildings is more complicated than doing so for 2D faces.

2.3 | Coloring sketches via self-referencing methods

Training cGANs for sketch-based architectural design is difficult because few datasets contain the necessary labels. Another difficulty is that different domains, such as the spatial layout, exterior style, and mechanical structure of buildings, are too heterogeneous to label within the same network. To overcome these inconveniences, many self-referencing methods have been developed, mainly based on two approaches:

2.3.1 | Introducing additional losses

Previously, style translation nets have used the content of one image and the style of another image, as expressed by convolutional features, to generate images without any labeling procedure (Gatys et al., 2016; Johnson et al., 2016). Based on these methods, a style loss can be added in the evaluation phase to reflect how well the style of the reference images is imitated, such as in the VGG-19 network (Simonyan & Zisserman, 2015).

Similarly, spatial information can also be encoded and controlled by adding losses. For example, Wang presented a method that enables flexible control of the output shapes and poses by utilizing an additional cross-domain adversarial loss to minimize the differences between three reference sketches and the sketch forms of the generated images (S. Y. Wang et al., 2021). However, this method cannot be used to customize other properties, such as color and texture, unless other types of losses are introduced.



2.3.2 | Multiheaded GANs

Generally, multiheaded GANs are mainly used for generation and fusion tasks, with specific mechanisms for each. The recently proposed Face Editing Generative Adversarial Network with User's Sketch and Color (SC-FEGAN) can generate a face image by superimposing red-green-blue (RGB) pixels, a sketch mask and incomplete images provided in nine channels as the inputs (Jo & Park, 2019). Similarly, if the color, shape, and other domains are encoded in multiheaded GANs, these domains can be controlled separately (Lee et al., 2020; S. Wu et al., 2022).

In Lee's research (Lee et al., 2020), the self-referencing method is used in multiheaded GANs to first disassemble a critical dataset into different subdatasets via specific operators. Then, the subdatasets are encoded by a multiheaded architecture and optimized based on corresponding losses. In this context, AMs (Vaswani et al., 2017), originally developed for semantic generation tasks, can also be used as connecting modules to integrate various subdatasets.

Notably, the losses utilized in Lee's method, such as the L1, content, and style losses, are all used for minimizing the differences between the input and predicted images during the training processes; this is fundamentally different from the approach of style-transforming models, which work by balancing the content and style losses (e.g., Li & Wand, 2016; X. Wang & Gupta, 2016). Based on these considerations, a mapping between content and color can be elaborately learned such that users can control the generated outcomes either by changing the input sketches provided as the basic image contents or by specifying particular reference images to indicate the desired color palette.

2.4 | Fusing sketches via GANs and DDIMs

Color images can be fused by interpolating their latent codes, as is done in the StyleGAN2 architecture (Karras et al., 2020). However, such methods overemphasize the integration of different domains and consequently cannot achieve sound results when fusing two sketches that easily contradict each other. In addition, the curved edge problem remains when integrating architectural sketches via GAN-based methods.

Recently, diffusion models have been shown to be superior to current state-of-the-art generative models (Ho et al., 2020; Song et al., 2020) for generating quality images. For example, a DDIM is trained by first generating images by adding noise to the data in an iterative manner via a particular Markovian diffusion process until mosaicking

is generated and then sampling via the reverse process. Notably, the results generated by a DDIM are more likely to follow the distribution of approximate cases than approximate the mean state of the entire dataset. Accordingly, the linear edges in architectural images can be effectively maintained by a DDIM.

3 | METHODOLOGY

3.1 | The Y-shaped GAN

3.1.1 | The two-headed encoder

Our proposed Y-shaped GAN generates a colorful architectural image I_g by integrating an input sketch I_s (as spatial information) and an input colorful image I_r (as color information). As illustrated in Figure 3, given a color image I (256×256 pixels) in our dataset, we first convert it into its corresponding sketch image I_s using the extended difference-of-Gaussians (XDoG; Winnemöller et al., 2012) operator. The Y-shaped GAN is based on Lee's GAN, but we delete the encoder that is used for encoding the distorted image. We utilize the image I as the input I_r without a twisting procedure. In Lee's method (Lee et al., 2020), twisted images must undergo a complicated transposition process using linear networks of AM modules to achieve the same purpose, and we suspect that this function is redundant. To maintain the triangular equilibrium state, we also delete the linear layers in the AM and the triplet loss.

Taking I_s and I_r as inputs, the Y-shaped GAN first encodes I_s into F_s and V_s and encodes I_r into F_r and V_r . Here, F_s and F_r denote the arrays of feature maps obtained after each convolution procedure, from the lowest layer (256×256) to the appropriate depth (128×128), to be used as spatial information. V_s and V_r are obtained by downsampling and concatenating these feature maps:

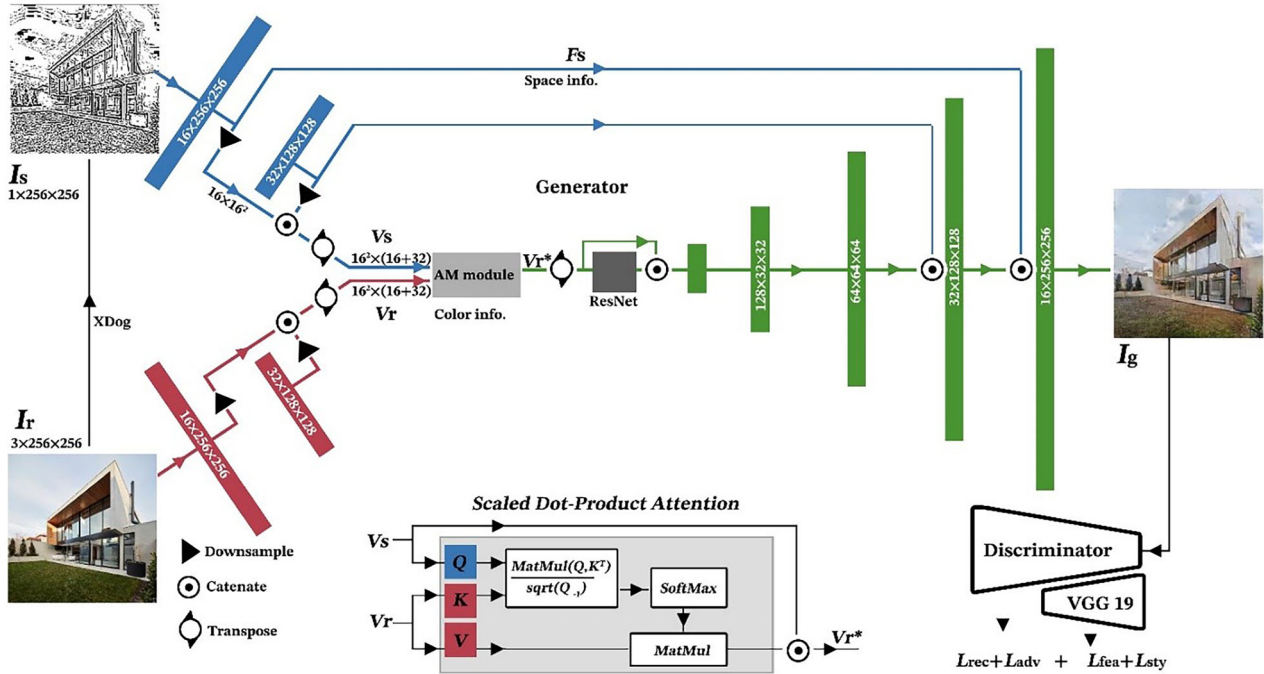
$$V = [\varphi(F^1); \varphi(F^2); \dots; \varphi(F^n)] \quad (1)$$

where F^n denotes the outputs of different layers, and φ denotes the downsampling function by which the different F^n are translated into dimensions of 16×16 pixels via the AvgPool2d method. Moreover, “;” denotes the feature-wise concatenation operator. Afterward, V_s and V_r should be transposed as follows:

$$V_{(Batch \times Feature \times 16 \times 16)} \rightarrow V_{(Batch \times 16^2 \times Feature)} \quad (2)$$

before being sent to the subsequent AM module.

(a) The Y-shaped sketch coloring GAN



(b) The U-Net in DDIM

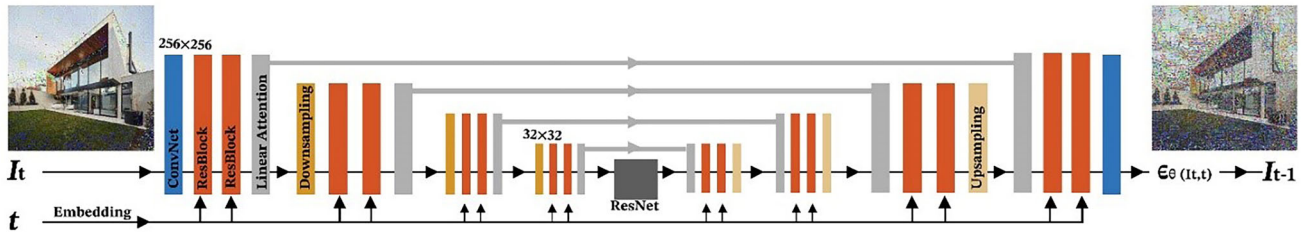


FIGURE 3 The overall workflows of the Y-shaped generative adversarial network (GAN) and the U-Net utilized in denoising diffusion implicit model (DDIM).

3.1.2 | AM module

Based on the scaled dot-product attention method (Vaswani et al., 2017), our AM module converts the color information V_r into a modified form V_r^* that “imitates” V_s . To realize this function, V_s is sent to the AM module as the query, while V_r is also sent to the AM module as both the key and the value (see Figure 3a); accordingly, the output V_r^* is calculated as follows:

$$\begin{aligned} V_r^* &= \text{Attention}(V_s, V_r, V_r) + V_s \\ &= \text{softmax}\left(\frac{V_s V_r^T}{\sqrt{d_s}}\right) V_r + V_s \end{aligned} \quad (3)$$

where T denotes transposition, and d_s is the dimensionality of V_s .

$$V_r^*_{(Batch \times 16 \times 2 \times Feature)} \rightarrow V_r^*_{(Batch \times Feature \times 16 \times 16)} \quad (4)$$

to obtain the final matrix for the subsequent decoding task.

The addition of V_s at the end of Equation (3) is important because V_s and F_s , as the starting and ending inputs of the decoder (see Figure 3), respectively, facilitate the matching between spatial and color information during the decoding process. If V_s were to be deleted, the generated results could not follow the changes in the reference image.

3.1.3 | The decoder

The AM module and decoder are connected by five ResNet layers (He et al., 2016), and the sketch encoder and decoder are bridged by skip connections. Accordingly, the final integrated output I_g is generated based on the matrix V_r^* and the spatial information F_s .

3.1.4 | The triangular equilibrium state mechanism

The key triangular equilibrium state is achieved by: (1) The feature maps of the color image are transposed in matrix



form in the AM module to delete their spatial information before being sent to the decoder; (2) the feature maps of the sketch are sent to the final layers of the decoder via skip connections. That is, a bifurcated flow for the sketch feature maps serves to provide guidance for the use of color information during the decoding processes (see Figure 3a).

3.1.5 | Objective function

L1 loss: Since the output image I_g simulates the ground-truth image I_r , we can directly impose a reconstruction loss to penalize the network for spatial and color differences as follows:

$$L_{rec} = \frac{1}{N} \sum_{i=1}^N \|I_{r,i} - G(I_{s,i}, I_{r,i})\|_1 \quad (5)$$

Adversarial loss: The discriminator D , as an adversary of the generator G , has the objective of distinguishing between I_g and I_r . The adversarial loss for optimizing D is formulated as follows:

$$L_{adv(d)} = \frac{1}{N} \sum_{i=1}^N [\log D(I_{r,i}) + \log(1 - D(G(I_{s,i}, I_{r,i})))] \quad (6)$$

The adversarial loss for optimizing G is:

$$L_{adv(g)} = \frac{1}{N} \sum_{i=1}^N [\log D(G(I_{s,i}, I_{r,i}))] \quad (7)$$

Perceptual losses: A style reconstruction loss can be used during training to penalize differences in colors, textures, and common patterns between I_g and I_r . We adopt the VGG-19 model pretrained on ImageNet (Simonyan & Zisserman, 2015) to achieve similarity in both content and style between I_g and I_r . The L1 loss between the feature representations is used as the feature reconstruction loss:

$$L_{fea} = \frac{1}{C} \sum_{l=1}^C \|\Phi_l(I_r) - \Phi_l(G(I_s, I_r))\|_1 \quad (8)$$

where L is the number of perceptual layers, and Φ_l represents the features from layer l in VGG-19. The style reconstruction loss is defined as the squared Frobenius norm of the distance between the Gram matrices of I_g and I_r :

$$L_{sty} = \frac{1}{C} \sum_{l=1}^C \|Gram_l(I_r) - Gram_l(G(I_s, I_r))\|_1 \quad (9)$$

where $Gram$ denotes the Gram matrix (Johnson et al., 2016). Finally, the objective function of the generator in the Y-shaped GAN is summarized as follows:

$$\min_G \max_D L_{total} = \lambda_r L_{rec} + \lambda_a L_{adv} + \lambda_f L_{fea} + \lambda_s L_{sty} \quad (10)$$

3.2 | The fusion method using a DDIM

3.2.1 | Training process and objective function

Following recent works (Ho et al., 2020; Song et al., 2020), we repeatedly train the U-Net-based DDIM on colorful architectural data (256×256 pixels) to minimize the following objective function:

$$Loss(t) = \mathbb{E}_{\epsilon, t, I_0} \left\| \epsilon - \epsilon_\theta \left(\sqrt{\alpha_t} I_0 + \sqrt{1 - \alpha_t} \epsilon, t \right) \right\|_2^2 \quad (11)$$

Here, $\epsilon \sim N(0, I)$. ϵ_θ is calculated by the U-Net-based estimator (see Figure 4b) from the in-process image I_t and the vector of positive coefficients $t \sim [1, 2, \dots, T]$ embedded in many revised residual blocks. I_0 is the initial input image. The fixed interval α_t is calculated based on t .

3.2.2 | Sampling and interpolation processes

Song et al. (2020) provided a standard sampling method for decoding noise I_T following a standard normal distribution and adding it to the image I_0 in an iterative manner via the following equation:

$$I_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{I_t - \sqrt{1 - \alpha_t} \epsilon_\theta(I_t, t)}{\sqrt{\alpha_t}} + \sqrt{1 - \alpha_{t-1}} \epsilon_\theta(I_t, t) \right) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(I_t, t) + \sigma_t \epsilon_t \quad (12)$$

where $\epsilon_\theta(I_t, t)$ is estimated by the trained U-Net. The random factor σ_t controls the sampling process in either a fixed ($\sigma_t = 0$) or random mode:

$$\sigma_t = \sigma_{\max} = \sqrt{(1 - \alpha_{t-1}) / (1 - \alpha_t)} \sqrt{(1 - \alpha_t) / \alpha_{t-1}} \quad (13)$$

Setting $\sigma_t = \sigma \times \sigma_{\max}$, we use $\sigma \sim [0, 0.2]$ for random generation, as shown in Figure 5, to achieve reasonable results.

However, for generating the target image I^f from the color-similar image pair $I_{t=0}^*$ and $I_{t=0}^{**}$ using a DDIM, Song did not provide an equation that could be used to encode $I_{t=0}$ into its latent vectors $I_t = T$. We use the

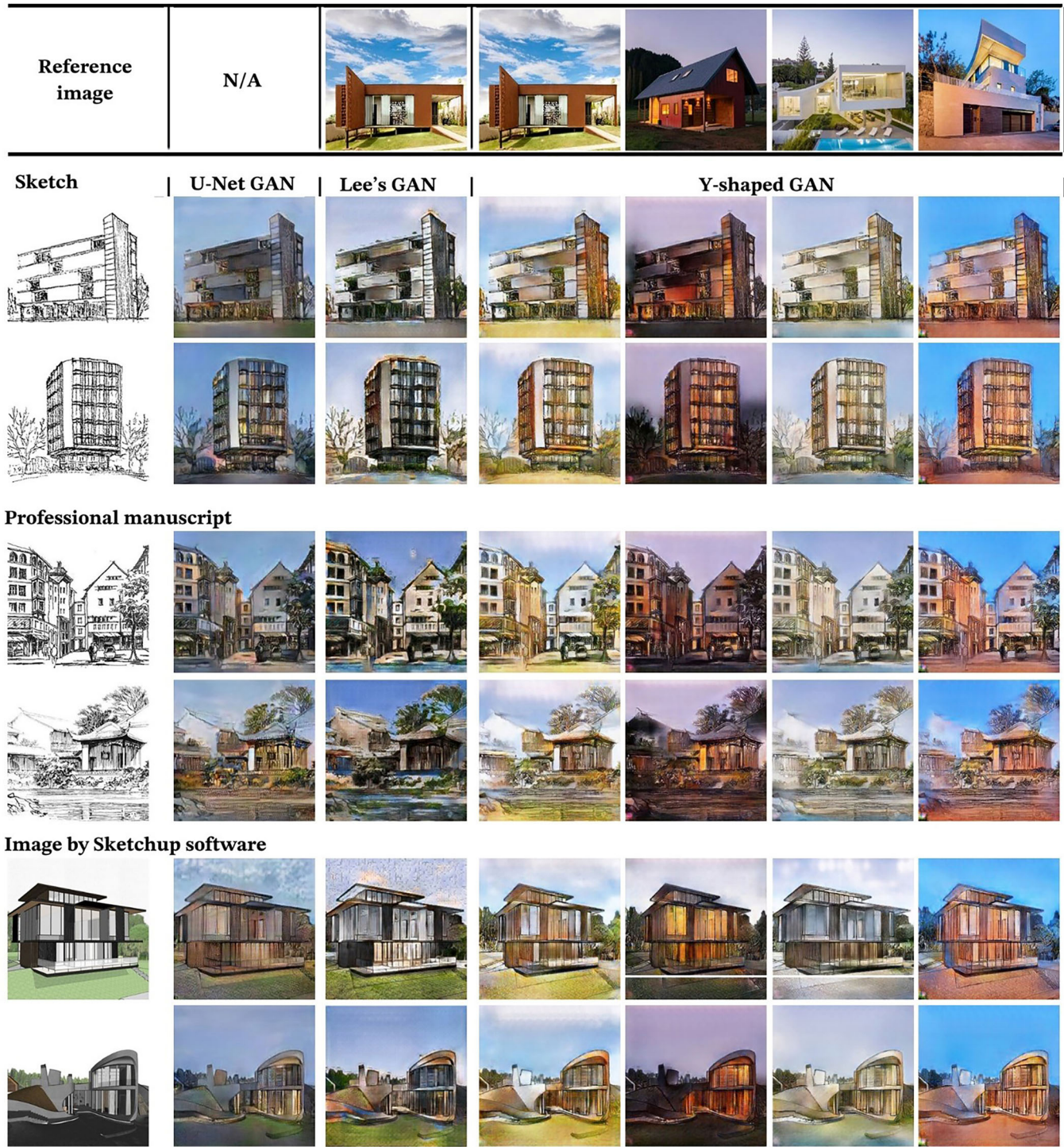


FIGURE 4 Coloring sketches via GANs.

following iterative equation to achieve this purpose:

$$I_{t+1} = \sqrt{\alpha_{t+1}} \left(\frac{I_t - \sqrt{1 - \alpha_t} \epsilon_{\theta}(I_t, t)}{\sqrt{\alpha_t}} + \sqrt{1 - \alpha_{t+1}} \epsilon_{\theta}(I_t, t) \right) \quad (14)$$

where the random factor σ_t is eliminated, meaning that the encoding process is deterministic.

Based on the above equations, given a pair of color-similar images $I_{t=0}^*$ and $I_{t=0}^{**}$, we first fuse them as follows:

$$I_{t=0} = I_{t=0}^* \times m + I_{t=0}^{**} \times (1 - m) \quad (15)$$

where m is the interpolation factor. Then, we encode the interim fuzzy image $I_{t=0}$ into its corresponding latent



(a) Coloring and fusing sketches



(b) Fusing sketch and reference image



(c) Comparison of different interpolating methods

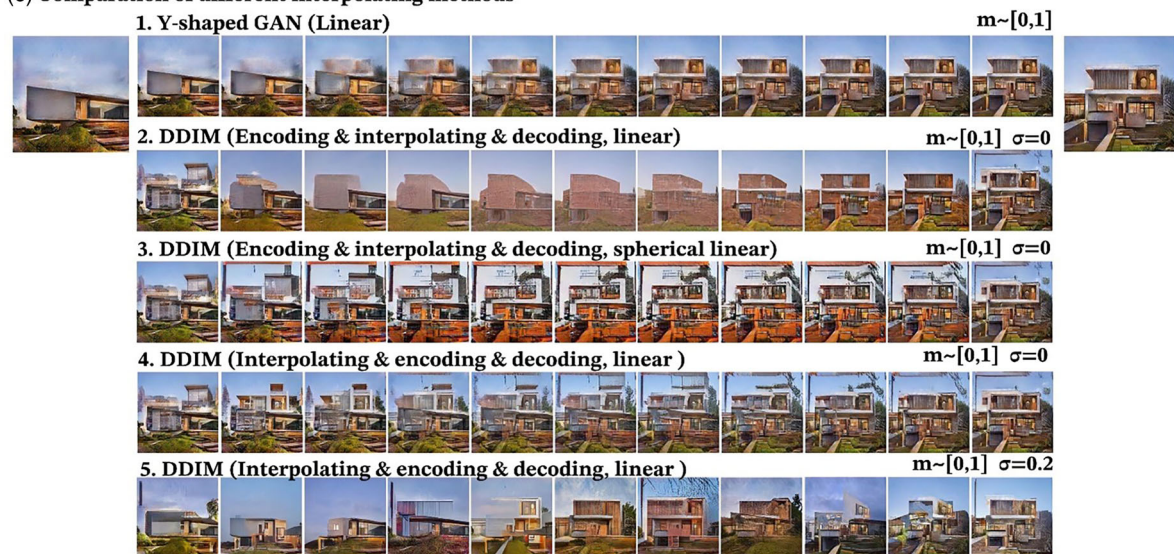


FIGURE 5 Fusing and coloring architectural sketches and images via Y-shaped GANs and DDIMs.



vectors $I_{t=T}$ using Equation (14). Subsequently, we decode the results using Equation (12).

For comparison, we also interpolate the latent vectors of the two images ($I_{t=T}^*$ and $I_{t=T}^{**}$) to obtain the fused latent vectors $I_{t=T}^f$ using the spherical linear method:

$$I_{t=T}^f = \frac{\sin[(1-m)\varphi]}{\sin\varphi} I_{t=T}^* + \frac{\sin[m\varphi]}{\sin\varphi} I_{t=T}^{**} \quad (16)$$

where m is the interpolation factor and

$$\varphi = \cos^{-1}(I_{t=T}^* \cdot I_{t=T}^{**}) \quad (17)$$

According to testing, this approach cannot achieve sound results (see Figure 5c).

3.3 | Data preparation and training details

We collected an architectural dataset to train our model. The architectural dataset consists of 10,000 images of low-rise buildings augmented from 600 initial photos collected from the Internet by common methods such as scaling, cutting, and mirror transition. These RGB images were resized to 256×256 pixels. As mentioned above, we then generated the corresponding sketch image dataset by means of the XDoG operator. When the trained GANs are applied, the sketches or images provided as input should be standardized using the same XDoG operator before being sent to the generator.

We randomly selected pairs of data, each consisting of a sketch and its corresponding color image, for training the Y-shaped GAN. We used the parameters of $\lambda_r = 20$, $\lambda_a = 1$, $\lambda_f = 0.01$, and $\lambda_s = 1$ for training; we note that these settings can be appropriately applied to a range of images without large absolute differences. We used the Adam solver for optimization, with a learning rate of 0.0002 for both G and D . The batch size was 1 in all of our training processes. The DDIM was trained on color architectural images using the default settings suggested by Song et al. (2020), such as $T = 1000$. It took 1 day to train the GANs and 5 days to train the DDIM using one 12-GB NVIDIA GeForce RTX 4070Ti GPU.

4 | RESULTS AND ANALYSIS

4.1 | Coloring sketches via the Y-shaped GAN

Figure 4 shows the overall qualitative results generated by our Y-shaped GAN model. Generally, this model correctly

matches the color information of the reference images with relevant spatial information from the given sketches when generating the façades, eaves, walls, and windows of buildings. The images generated via our Y-shaped GAN method can also imitate the overall environments of the reference images, such as the sky, grass, water, and illumination. The color palettes of specified reference images with different atmospheres, such as “secluded,” “springy,” “coquettish,” and “pastoral,” can also be imitated.

We performed different tests in which the elementary sketches, the professional hand-drawn sketches, and the sketch images created by SketchUp software were used as inputs for generation. These sketches were standardized by means of the XDoG operator before being sent to the Y-shaped GAN generator. These results indicate that colorful architectural images can be generated from these types of inputs. They also indicate that our methods can effectively assist in real-world architectural design tasks.

We compare our method with the U-Net based GAN architecture and Lee’s GAN. The U-Net based GAN method generates images with only a single style obtained by balancing every case in the dataset. The initial version of Lee’s GAN cannot utilize input reference images for architectural design. However, by deleting the encoder used for encoding distorted images, the linear layers in the AM, and the triplet loss, our Y-shaped GAN can effectively utilize the reference images.

4.2 | Fusing sketches and images via the Y-shaped GAN and DDIM

The images in the right column of Figure 5a show the color-similar intermediate images obtained using the Y-shaped GAN and the subsequent fusion results obtained using the DDIM by fusion, encoding via the fixed mode and decoding using a random factor σ . Our method can also be used for fusing a sketch and a reference image if taking the XDoG image of the reference image as the second sketch (see Figure 5b).

The first advantage of this combined method is that the fusion results do not depend entirely on the spatial positions extracted from the input sketches. That is, our method can integrate two architectural sketches using the potential knowledge learned by the network and automatically utilized during the sampling processes instead of performing fusion via the copy-and-paste technique. Second, the fused images generated by the DDIM can preserve the linear, rectangular, and gravity-based structural characteristics of buildings regardless of what differences exist between the input sketches.

Two disadvantages are observed when using this combined method. First, it is worth noting that the sampling



time required to generate one fused image with DDIM is more than 1 mine. Second, although one excellent design is sufficient for architects to complete their work, we estimate that only approximately 50% of the generated architectural images are subjectively reasonable even if σ is set appropriately. We selected relatively reasonable results for presentation in Figure 5.

4.3 | Evaluation results

We compare our methods against recently developed deep learning approaches in terms of two types of metrics based on our architectural datasets.

4.3.1 | Fréchet inception distance (FID)

The FID is a well-known metric that can evaluate the performance of models by measuring the Wasserstein-2 distance between the featural representations of real images and the corresponding generated outputs (Heusel et al., 2017). A low FID score indicates that the model generates images with quality and diversity close to those of the real data distribution.

4.3.2 | Multiscale structural similarity index measure (MS-SSIM)

The MS-SSIM is a measure of structural similarity as an approximation index between inputs and outputs (Z. Wang et al., 2003). MS-SSIM scores are between 0 and 1. The closer to 1 the score is, the more structural information of the input sketches is preserved by the generated results. Accordingly, we mainly use this metric to evaluate the fusion performance using the following equation:

$$M = M(A, G) \times 0.5 + M(B, G) \times 0.5 \quad (18)$$

where A and B are input sketches, and G is the fused image. When evaluating the coloring tasks, A is the input sketch, and B is the reference image.

The evaluation results are shown in Table 1. For coloring tasks, our Y-shaped GAN produces a lower FID score than Lee's GAN, indicating that our method can better color architectural sketches and generate more realistic outputs. The score of the Y-shaped GAN is slightly higher than that of the U-Net GAN. However, the coloring processes are more controllable than those of the W-GAN, so the MS-SSIM score is correspondingly higher.

For fusion tasks, the method of combining the Y-shaped GAN and a DDIM achieves the lowest FID and highest

TABLE 1 Evaluation results.

		FID	MS-SSIM
Coloring sketches			
1. W-GAN		48.81	0.28
2. Lee's GAN		83.65	0.48
4. Y-shaped GAN		56.43	0.32
Fusing sketches			
1. Lee's GAN		124.64	0.20
2. Y-shaped GAN		92.28	0.57
3. U-Net + Y-shaped GAN		88.05	0.45
4. Y-shaped GAN + DDIM			
4-1. Performing encoding,	$\sigma = 0.0$	158.62	0.40
interpolation and decoding	$\sigma = 0.1$	111.61	0.39
(spherical linear)	$\sigma = 0.2$	92.28	0.36
4-2. Performing interpolation,	$\sigma = 0.0$	95.89	0.57
encoding and decoding	$\sigma = 0.1$	56.35	0.51
(linear)	$\sigma = 0.2$	60.82	0.43

Note: A lower FID score is better; a higher MS-SSIM score is better.

Abbreviations: DDIM, denoising diffusion implicit model; FID, Fréchet inception distance; GAN, generative adversarial network; MS-SSIM, multiscale structural similarity index measure.

MS-SSIM scores in comparison with the other methods by following the sequence of interpolation, encoding, and decoding. Rather, the DDIM tends to generate meaningless intermediate states if the encoding is performed before interpolation, which can be seen in Figure 5c and is clarified in Table 1(4-2). We suppose that the series of the interpolation, encoding, and decoding processes in the DDIM can be regarded as a type of search that seeks the nearest distribution to optimize the fuzzy intermediate images that are interpolated by color-similar images.

According to our testing results, the random factor $\sigma = 0.1$ is a preferable setting. First, it yields interesting fusion results; second, because the intermediate images generated by the Y-shaped GAN are novel cases that differ from the distribution of the training data, the DDIM usually generates chaotic results if we set $\sigma = 0$ (see Figure 5c-3) with high FID scores. Setting $\sigma = 0.1$ can remove these blurs. If we set $\sigma > 0.2$, the network only generates nearly random images, so it fails to preserve the contents of the input sketches, and the resulting MS-SSIM scores will be low.

5 | CONCLUSION AND FUTURE WORK

In this study, we consider methods for coloring and fusing architectural sketches and for generating the final results of sketch-based design. (1) For coloring tasks, we integrate the content of sketch A with the color information of image B via a Y-shaped GAN. Our Y-shaped GAN can effectively



color sketches by exploiting an AM module and maintaining a triangular equilibrium state. We test this method on coloring both hand-drawn sketches and sketches created with 3D software, and the results indicate that they can be reasonably colored by imitating the reference images than existing methods. (2) For coloring and fusion tasks, we first generate color-similar image pairs by a Y-shaped GAN and fuse them via the DDIM. This approach addresses the basic question of how to translate the incompatible states of two architectural structures into a single globally optimal state through deep learning. We provide evidence that the novel method combining a Y-shaped GAN and a DDIM can effectively fuse sketches or images into a final harmonious design and is better than the existing methods. They are also the critical evidence that the two basic working patterns including (1) balancing contradictory design cases and (2) conceiving design works using references can be effectively enhanced by our novel deep learning methods.

In this research, we also explore the approach that utilizes reference cases for one-shot coloring and fusing. This approach works based on the prerequisite that the target reference images are excellent in terms of certain aspects or at least better than the uncompleted sketches. Although the aesthetic goal is emphasized here more than the classic goals, such as weights and costs, we believe that the new method is an eclectic but potential approach for linking the conceptual and structural design processes to avoid splitting them.

Our methods are restricted to the spatial and color domains though other types of information are involved finitely. The first future work task will be to extract and utilize these implicit information for fusing sketches by possible methods. The second future work task will be to link our method with long-lasting structure optimization studies by embedding expert knowledge to direct the generation processes or by utilizing optimization methods such as neural dynamics algorithms (Park & Adeli, 1997) and neural dynamic classification algorithms (Rafiei & Adeli, 2017). The third future work will be to utilize more powerful supervised machine learning algorithms such as dynamic ensemble learning algorithms (Alam et al., 2020), finite-element mechanisms (Pereira et al., 2020), and self-supervised learning (Rafiei et al., 2022) to improve the efficiency and quality of our methods.

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